



Sir Syed University of Engineering & Technology, Karachi

CS-468

Software Quality Assurance & Testing

Week 1

Session 1

Course Overview

CS – 468

Software Quality Assurance & Testing
(3 credit hour)

Pre-req: Software Engineering

... Course Overview ...

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... Course Overview ...

Course Outline

- Software error fault and failure, Causes of Software errors,
- Software quality factors
- Software Process and Process Improvement, Models, Standards, Guidelines, Capability Maturity Model (CMM), ISO software quality standard
- Introduction to testing, principles of testing, fundamental test process, Test cases, Testing and test cases design techniques, The psychology of testing, Code of ethics
- Test Management, Incident Management, Configuration Management, Introduction to automated mode of testing, Testing Maturity Model, Defect and experienced based testing

... Course Overview ...

Course Assessment Methods:

• Quizzes/Assignments	20%
• Midterm Exam	30%
• Final Exam	50%
• Total	100 Marks

... Course Overview ...

TEXT BOOK:

- Daniel Galin, **Software Quality Assurance From theory to implementation**, Pearson, Latest Edition.
- Brian Hambling, **Software Testing, ISTQB–ISEB Foundation Guide**, Latest Edition

REFERENCE BOOKS:

- Kshirasagar Naik, Priyadarshi Tripathy, **Software Testing and Quality Assurance – Theory and Practice**, Wiley, Latest Edition
- Rex Black, **Advanced Software Testing Guide to the ISTQB Advanced Certification as an Advanced Technical Test Analyst**, ISTQB, Latest Edition

... Course Overview

- Course Information Sheet (CIS)

Introduction

- **Software**, from the SQA perspective, is the combination of computer programs (the “code”), procedures, documentation, and data necessary for operating the software system. The combination of all four components is needed to assure the quality of the development process as well as the ensuing long years of maintenance.

... Introduction ...

- **Software quality**, according to Pressman's definition, is the degree of conformance to specific functional requirements, specified software quality standards, and Good Software Engineering Practices (GSEP).

... Introduction ...

- **Software quality assurance:** According to the IEEE expanded definition, software quality assurance is the systematic, planned set of actions necessary to provide adequate confidence that a software development or maintenance process conforms to established functional technical requirements as well as the managerial requirements of keeping to schedules and operating within the budget.

... Introduction ...

- **Objectives of SQA activities for software development and maintenance are:**
 - Assuring, with acceptable levels of confidence, conformance to functional technical requirements.
 - Assuring, with acceptable levels of confidence, conformance to managerial requirements of scheduling and budgets.
 - Initiating and managing activities for the improvement and greater efficiency of software development and SQA activities.

... Introduction ...

- **Relationship between software quality assurance and software engineering:**
 - Software engineering is the application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software.
 - The characteristics of software engineering, especially its systematic, disciplined and quantitative approach, make software engineering a good environment for achieving SQA objectives.
 - It is commonly accepted that cooperation between software engineers and the SQA team is the way to achieve efficient and economic development and maintenance activities that, at the same time, assure the quality of the products of these activities.